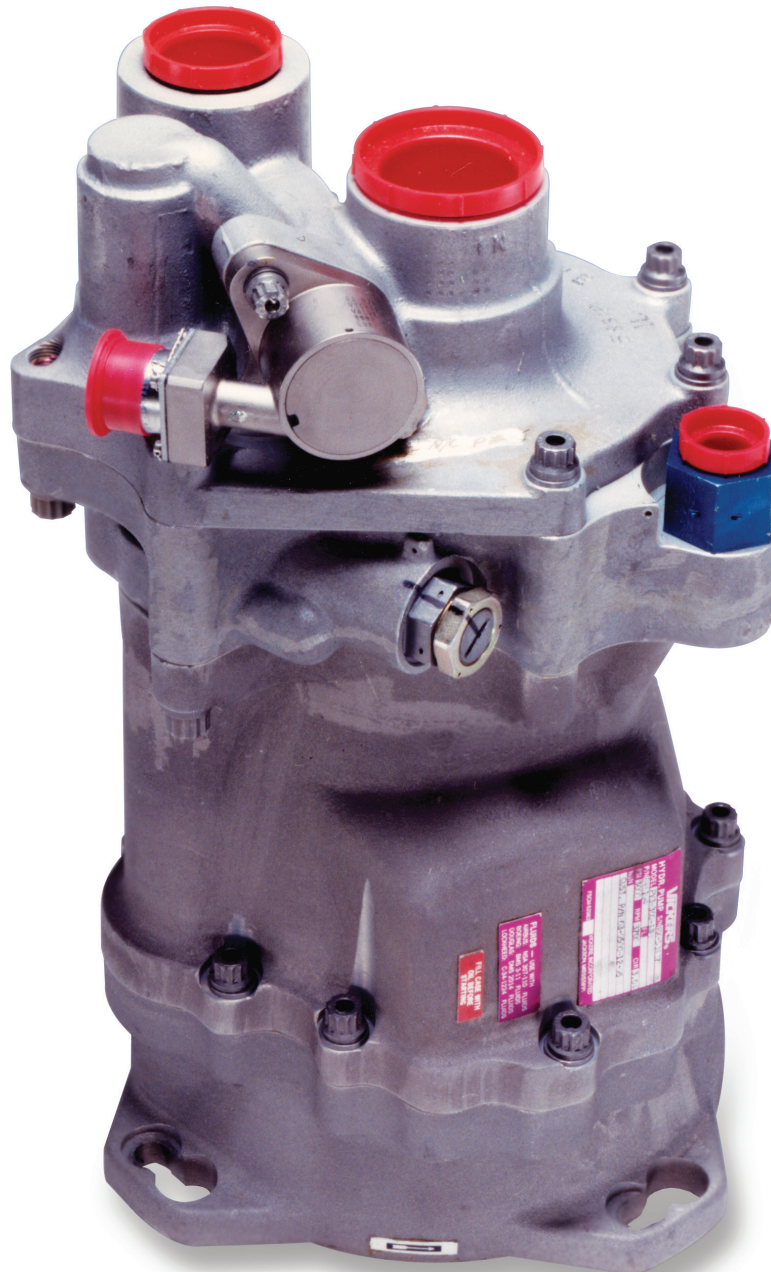






# Vickers Engine-Driven Pump on the Airbus A340 Aircraft P/N 972630 (Model PV3-300-13F)

Eaton's Model PV3-300-13F (P/N 972630) Engine-Driven Pump (EDP) was developed for use on the Airbus A330 and A340 commercial aircraft. The design of this pump is based on the proven PV3-240 series pumps that have flown millions of hours on Airbus, Boeing, Douglas and Lockheed commercial airliners which incorporate numerous improvements that allow the unit to achieve a mean-time-between-failure of 20,000 hours.

## DESIGN FEATURES

The Model PV3-300-13F is a 46 gpm (174 L/min), 3000 psi (20685 kPa) hydraulic pump that incorporates numerous design features to enhance the reliability and maintainability of the unit.

### **Increased Fatigue Strength**

Structural design changes and use of improved casting materials were incorporated into the pump.

### **Improved Shaft and Yoke Bearings**

All bearings used in the PV3-300-13F model EDP are manufactured from consumable electrode vacuum melted (CEVM) M-50 bearing quality tool steel.

### **Improved Cylinder Block Material**

The cylinder block is machined from ductile iron to minimize wear in the drive spline and piston bores. Bronze plating on the cylinder block face provides a durable bearing surface for wear against the tool steel wafer plate.

### **Yoke**

The yoke has been stiffened to minimize deflections under the piston shoes. Additionally, the yoke O.D. has been designed to allow installation of mid-grip

helical thread locking inserts. This eliminates the need for time consuming safety wiring of the bearing shoe hold down plate threaded fasteners.

### **Balanced Blocking Valve**

The outlet blocking valve is hydraulically balanced for rapid response and ease of manufacture. Viscous dampening of the blocking valve piston retards the closure rate to allow sufficient decompression of the system outlet fluid prior to valve closure.

### **Flange Mounted Electrical Depressurization Valve (EDV)**

The EDV is flange mounted using two (2) threaded fasteners. This design ensures consistent positioning of the electrical connector master slot and reduces the possibility of handling damage during installation.

### **Pressure Compensator**

The pressure compensator is located in the adaptor block thereby eliminating a high pressure parting line static seal. The pressure compensator design also incorporates viscous dampening to improve dynamic stability.

### **Optimum Flange Bolt Patterns**

All bolted flange interfaces have been designed for optimum bolt loading and minimal deflection.

### **Stepped Valve Bushings**

The pressure compensator and EDV utilize stepped diameter sealing glands to minimize seal damage during assembly.

## BASIC PUMP OPERATION

The aircraft's engine rotates the pump drive shaft and the connected cylinder block and pistons. Pumping action is generated by piston shoes

which are restrained and slide on the shoe bearing plate in the yoke assembly. Because the yoke is at an angle to the drive shaft, the rotary motion of the shaft is converted to piston reciprocating motion.

As the piston begins to withdraw from the cylinder block, system inlet pressure forces fluid through a porting arrangement in the valve plate into the cylinder bore. The piston shoes are restrained in the yoke by a hold-down plate and hold-down retainer during the intake stroke. As the drive shaft continues to turn the cylinder block, the piston shoe continues following the yoke bearing surface. This begins to return the piston into its bore, toward the valve block.

The fluid contained in the bore is pre-compressed then expelled through the valve block outlet port. Discharge pressure holds the piston shoe against the yoke bearing surface during the discharge stroke and also provides the shoe pressure balance and fluid film through an orifice in the piston and shoe sub-assembly.

With each revolution of the drive shaft and cylinder block, each piston goes through the pumping cycle described above, completing one intake and one discharge stroke. High-pressure fluid is ported out through the valve block, past the blocking valve, to the pump outlet. The blocking valve is designed to open at approximately 400 psi (2758 kPa) outlet pressure and remains open during normal pump operation.

Internal leakage keeps the pump housing filled with fluid for lubrication of rotating parts and cooling. The leakage is returned to the system

through a case drain port. The case relief valve protects the pump against excessive case pressure, relieving it to the pump inlet.

## CONTROL FEATURES

### **Normal Pumping Mode**

The pressure compensator is a spool valve that is held in the closed position by an adjustable spring load. When pump outlet pressure (system pressure) exceeds the pressure setting, 2850 psi (19650.75 kPa), the spool moves to admit fluid from the pump outlet into the actuator piston. (In the schematic, the pressure compensator is shown at cracking pressure; i.e., pump outlet pressure just high enough to move the spool to begin to admit fluid to the actuator piston.)

The yoke is supported inside the pump housing on two bearings. At pump outlet pressures below 2850 psi (19650.75 kPa), it is held at its maximum angle - in relation with the drive shaft center-line - by the force of the yoke return spring. Decreasing system flow demand causes outlet pressure to become high enough to crack the compensator valve open and admit fluid to the actuator piston.

This control pressure overcomes the yoke return spring force and strokes the pump yoke to a reduced angle. The reduced angle of the yoke results in a shorter stroke for the pistons and reduced displacement. The lower displacement results in a corresponding reduction in pump flow. The pump delivers only that flow required to maintain the desired pressure in the system. When there is no demand for flow from

the system, the yoke angle decreases to nearly zero degrees stroke angle. In this mode, the unit pumps only its own internal leakage.

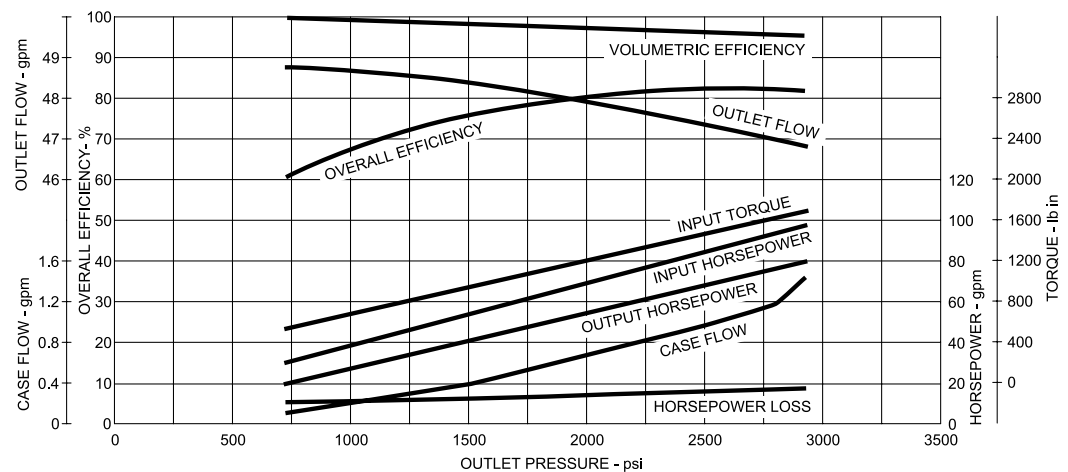
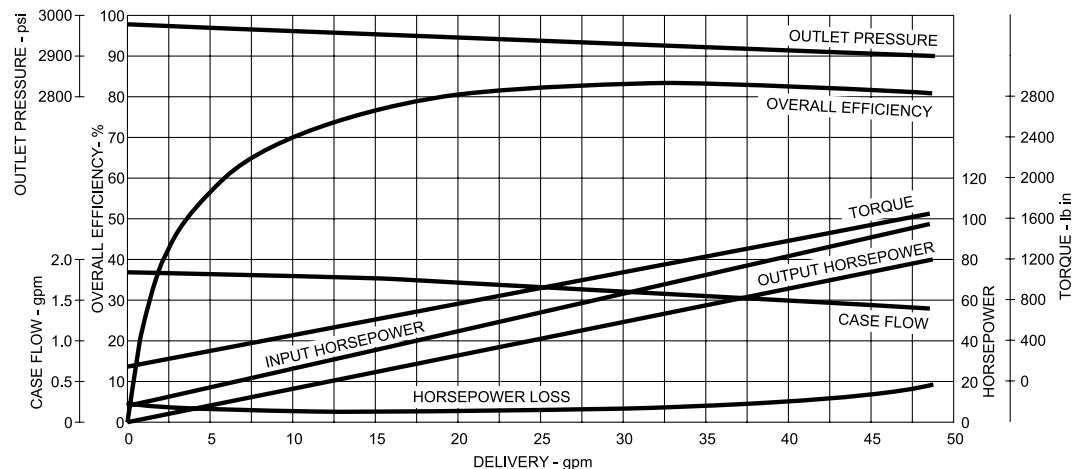
Thus, at pump outlet pressures above 2850 psi (19650.75 kPa), pump displacement decreases as outlet pressure rises. At system pressures below this level, no fluid is admitted through the pressure compensator valve to the actuator piston and the pump remains at full displacement, delivering full flow. Pressure is then determined by the system demand.

### DEPRESSURIZED MODE

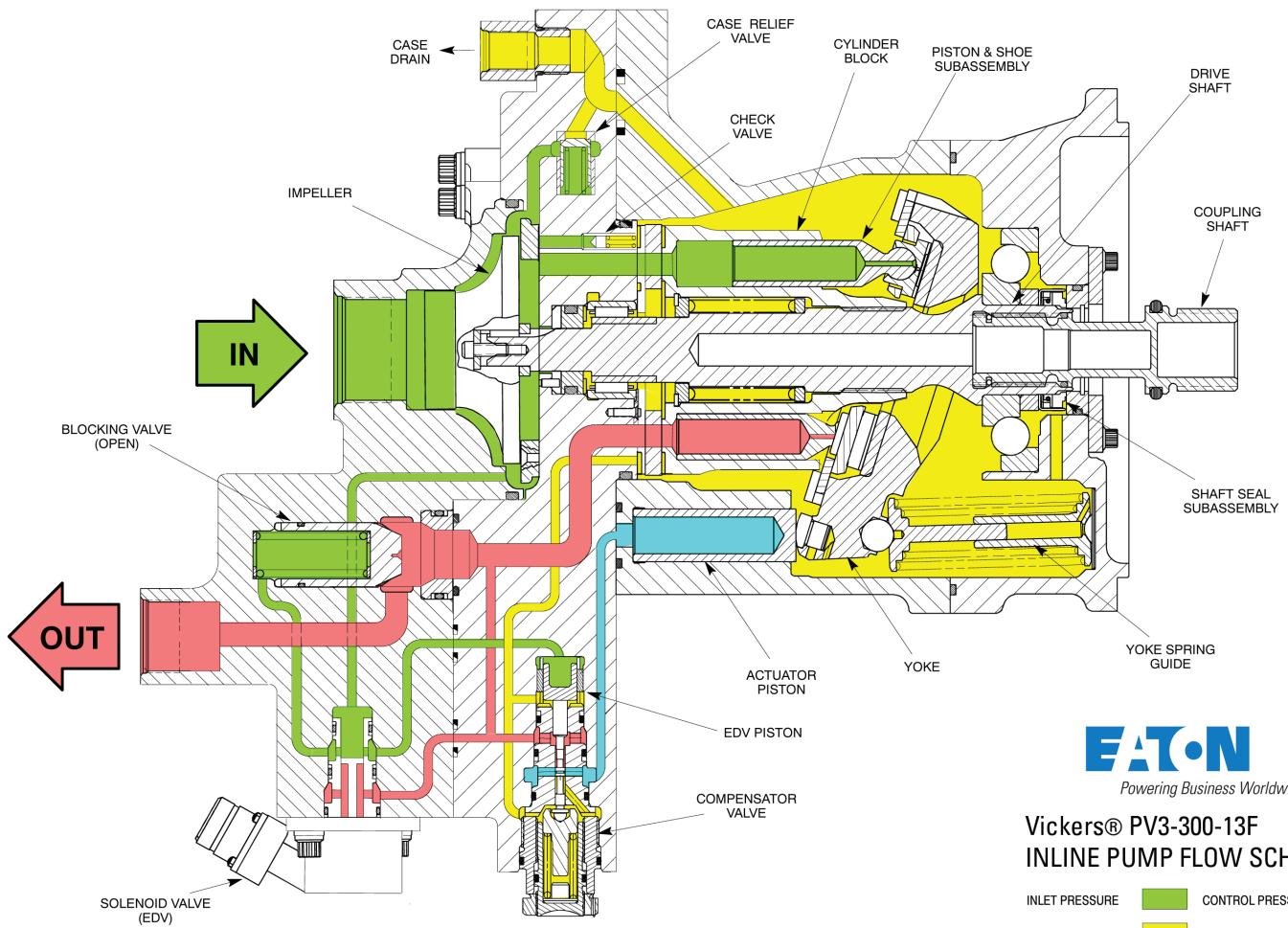
When the solenoid valve is energized, outlet fluid is ported to the EDV control piston on the end of the compensator. The high pressure fluid pushes the compensator spool beyond its normal metering position. This removes the compensator from the circuit and connects the actuator piston directly to the pump outlet. Outlet fluid is also ported to the blocking valve spring chamber. This equalizes pressure on both sides of its plungers, and the blocking valve closes due to the force of the blocking valve spring, and isolates the pump from the external hydraulic system. The pump strokes itself to zero delivery at an outlet pressure equal to the pressure required on the actuator piston to reduce the yoke angle to nearly zero. This depressurization and blocking feature can be used to reduce the load on the engine during starting and, in a multiple pump system, to isolate one system for check out purposes.

## Typical Pump Characteristics

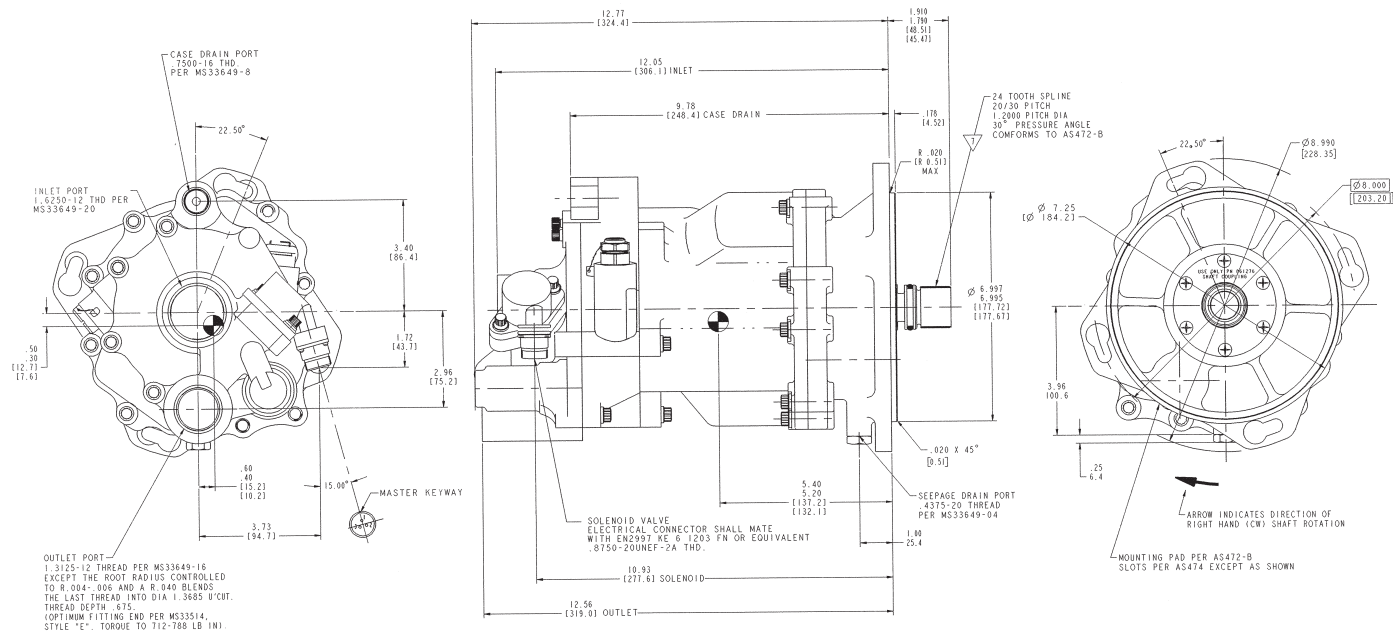
A representative speed of 3700rpm was used to develop curves



# Inline Pump Flow Schematic



# PV3-160-4/4B



Dimensions: inches (mm)

Eaton reserves the right to revise/modify information contained herein.



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